Assignment 1: Time series datasets

Source snapshots retrieved 5 October 2026

Weekly, monthly and quarterly data; at least 100 observations each. The two market datasets are multivariate; European macroeconomics adds a different domain. All exported attributes have zero missing values. Calendar months and quarters are regular periods despite differing day counts.

1. NVIDIA stock

Weekly | Individual technology stock

CSV: `01_nvda_weekly.csv`
Frequency: Weekly **Observations:** 312 **Attributes:** 3 (including date)
Time span: 2020-10-16 to 2026-10-02 (Friday labels)
Underlying trading dates: 2020-10-12 to 2026-10-02.

Sources.

Yahoo Finance: NVDA historical data `https://finance.yahoo.com/quote/NVDA/history/`

Attributes and missing data.

`date`: Friday labelling a Saturday-Friday week; YYYY-MM-DD. Missing: 0.

`nvda_adjusted_close_usd`: Last trading day's adjusted close in the week, USD per share. Yahoo adjusts this for splits and dividends. Missing: 0.

`nvda_volume_shares`: Sum of the daily trading volumes reported by Yahoo within the week, in shares. Missing: 0.

Numeric summary.

Attribute	Min	Max	Mean	Std. dev.
`nvda_adjusted_close_usd`	11.19	233.95	82.12	70.38
`nvda_volume_shares`	400,749,200.00	4,308,297,000.00	1,736,870,065.38	815,917,634.39

Sample standard deviation ($n-1$); calculated from exported CSV values. Display rounded to two decimals.

Reduction and preparation.

Daily observations were aggregated into Saturday-Friday weeks: last adjusted close and sum of volume. Only the most recent 312 complete weeks were retained. Open, high, low and unadjusted close were removed. A holiday Friday keeps the last actual quote within that week; the date is a period label. No prices were interpolated or forward-filled.

Why this series is useful.

An individual stock with strong trends and changing trading activity. Useful for studying price versus volume and the effect of differencing or returns on a trending series.



2. S&P 500 and VIX

Monthly | Equity market level and expected volatility

CSV: `02_sp500_vix_monthly.csv`
Frequency: Monthly **Observations:** 108 **Attributes:** 4 (including date)
Time span: 2017-10-01 to 2026-09-01 (month-start labels)
Covered calendar months: 2017-10 through 2026-09, ending 2026-09-30.

Sources.

Yahoo Finance: S&P 500 historical data `https://finance.yahoo.com/quote/%5EGSPC/history/`

Cboe: VIX historical data `https://www.cboe.com/tradable_products/vix/vix_historical_data`

Attributes and missing data.

`date`: First day labelling the calendar month; YYYY-MM-DD. Missing: 0.

`sp500_mean_close`: Arithmetic mean of the daily S&P 500 closes, in index points; a price index excluding dividends. Missing: 0.

`vix_mean_close`: Arithmetic mean of daily VIX closes, in volatility index points (annualized percentage units). Missing: 0.

`vix_max_close`: Largest daily VIX close in the month, in the same units; this is not an intraday high. Missing: 0.

Numeric summary.

Attribute	Min	Max	Mean	Std. dev.
`sp500_mean_close`	2,557.00	7,711.32	4,360.69	1,436.13
`vix_mean_close`	10.13	57.74	19.35	6.62
`vix_max_close`	11.30	82.69	24.51	10.14

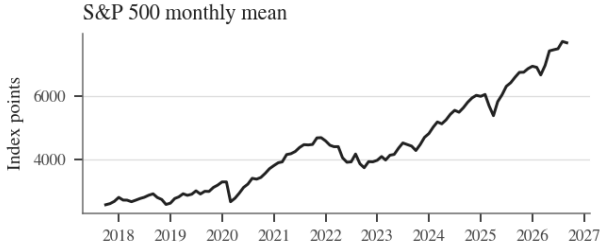
Sample standard deviation ($n-1$); calculated from exported CSV values. Display rounded to two decimals.

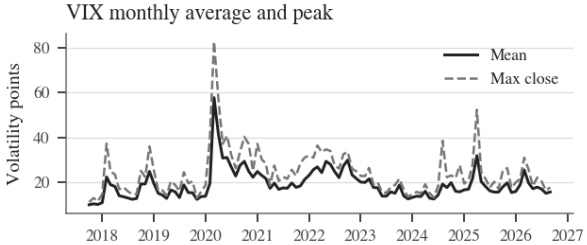
Reduction and preparation.

Daily closes were grouped by calendar month: mean for each index and maximum daily close for VIX. The most recent 108 complete months were retained; incomplete October 2026 was excluded. Within this window, 33 extra VIX holiday observations were excluded to use the same S&P 500 trading dates for both indexes. Other daily attributes were removed. No missing values were filled.

Why this series is useful.

A multivariate dataset contrasting broad equity-market trends with volatility spikes. The maximum VIX close retains information about stress that a monthly average can smooth out. No relationship between the variables is assumed.





3. Euro-area real GDP index

Quarterly | European macroeconomics

CSV: `03_euro_area_real_gdp_quarterly.csv`
Frequency: Quarterly **Observations:** 120 **Attributes:** 2 (including date)
Time span: 1996-07-01 to 2026-04-01 (quarter-start labels)
Covered quarters: 1996Q3 through 2026Q2, ending 2026-06-30.

Sources.

Eurostat: quarterly GDP (namq_10_gdp) `https://ec.europa.eu/eurostat/databrowser/view/namq_10_gdp/default/table?lang=en`

Attributes and missing data.

`date`: First day labelling the quarter (January, April, July or October); YYYY-MM-DD. Missing: 0.

`real_gdp_index`: Real GDP in chain-linked volumes, index 2020 = 100; seasonally and calendar adjusted. Covers the fixed 20-country euro-area group (EA20). Measures economic output excluding inflation. Missing: 0.

Numeric summary.

Attribute	Min	Max	Mean	Std. dev.
`real_gdp_index`	74.33	114.67	95.81	10.51

Sample standard deviation ($n - 1$); calculated from exported CSV values. Display rounded to two decimals.

Reduction and preparation.

Eurostat filters: frequency Q; geography EA20; GDP B1GQ; unit CLV_I20; adjustment SCA. The selected source contains 126 quarters from 1995 Q1 through 2026 Q2; only the latest 120 were retained, removing the oldest six. Quarter codes were converted to quarter-start dates. Constant series metadata were omitted from the CSV. Original index values were preserved without aggregation or imputation.

Why this series is useful.

Adds a European economic domain and slower frequency. Useful for studying long-term growth, recessions and shocks such as the COVID-19 contraction and subsequent rebound. The source's seasonal and calendar adjustment helps focus on economic cycles.

